

Predictive Forecasting of Care Load & Placement Demand

Research Paper: Exploratory Data Analysis, Forecasting Insights, and Recommendations

Project context: HHS Unaccompanied Alien Children (UAC) Program

Abstract

This study develops a predictive forecasting framework for estimating future children in HHS care using historical operational data. The workflow combines data cleaning, exploratory data analysis, feature engineering, baseline forecasting methods, machine-learning models, multi-horizon forecasting, empirical uncertainty intervals, and capacity-risk analysis. The final Gradient Boosting model produced substantially lower forecast error than the tested baseline and statistical approaches, particularly for short-term horizons. Results show that 1-day and 7-day forecasts are materially more reliable than 14-day and 30-day forecasts, highlighting the importance of uncertainty-aware planning rather than relying only on point forecasts.

1. Objectives

The project aimed to: (1) understand historical care-load and placement-related patterns; (2) identify useful operational and temporal predictors; (3) compare baseline, statistical, and machine-learning forecasting approaches; (4) produce 1-day, 7-day, 14-day, and 30-day forecasts; (5) quantify forecast uncertainty; and (6) translate forecasts into a practical capacity-risk signal for decision support.

2. Data and Data Quality

The dataset covered 12 January 2023 through 21 December 2025. After removing fully blank rows, 720 observations remained. The expected daily date range contained 1,075 dates, leaving 355 dates without observations. Missingness was highly systematic: Friday and Saturday accounted for most missing dates. This means the source is not a simple complete daily time series, and reporting cadence must be considered when interpreting temporal patterns and model results.

3. Exploratory Data Analysis and Feature Engineering

The analysis examined children apprehended and placed in CBP custody, children in CBP custody, children transferred out of CBP custody, children in HHS care, and children discharged from HHS care. Calendar features included day of week, day of month, month, quarter, year, week of year, and weekend indicators. Additional predictors included net pressure, discharge-to-transfer ratio, lagged operational measures, and rolling statistics. These features were designed to capture recent care-load conditions, operational flow, and recurring temporal patterns.

4. Model Evaluation

Baseline and statistical approaches included a naive forecast, 7-day moving average, ARIMA(1,1,1), and exponential smoothing. Gradient Boosting and Random Forest models were also evaluated. Gradient Boosting was the strongest overall machine-learning approach. For the final multi-horizon evaluation, Gradient Boosting achieved the following results:

Horizon	MAE	RMSE	MAPE
1-Day	81.73	101.76	3.74%
7-Day	145.66	170.93	6.74%
14-Day	344.23	389.87	16.00%
30-Day	365.18	404.04	16.56%

5. Forecasting Insights

Forecast performance decreases as the prediction horizon increases. The 1-day forecast has a MAPE of approximately 3.74%, while the 7-day forecast remains comparatively strong at 6.74%. Error increases materially at 14 and 30 days, with MAPEs of approximately 16.00% and 16.56%. Longer-horizon forecasts also show a positive prediction bias: the model tends to overpredict care load, with mean bias of about -13.81% at 14 days and -10.36% at 30 days when bias is expressed relative to mean actual values. This suggests that longer-range outputs should be used as planning signals rather than precise point estimates.

6. Uncertainty and Capacity Risk

Prediction intervals were estimated from empirical residual quantiles. The resulting approximately 95% intervals achieved retrospective coverage of about 93.9% to 94.8% across the four horizons. A scenario capacity threshold of 3,000 children was then used to illustrate potential capacity pressure. Retrospective empirical mean breach risk was 0% for 1-day and 7-day forecasts, approximately 0.14% for 14-day forecasts, and approximately 2.07% for 30-day forecasts. The maximum observed empirical breach-risk estimate reached about 4.12% for 14 days and 22.68% for 30 days. These are diagnostic scenario estimates, not calibrated real-time probabilities.

7. Recommendations for Government Stakeholders

1. **Prioritize short-term forecasts for operational decisions.** The 1-day and 7-day horizons provide the strongest predictive performance and are most appropriate for near-term staffing, placement, and resource coordination.
2. **Use longer horizons for scenario planning.** 14-day and 30-day forecasts should support preparedness discussions rather than be treated as exact demand estimates.
3. **Present prediction intervals alongside point forecasts.** Decision-makers should see a range of plausible demand, especially when evaluating capacity thresholds.
4. **Monitor reporting completeness.** The systematic Friday/Saturday gaps in the source data can affect trend analysis and model development. A consistent daily reporting process would improve forecasting reliability.
5. **Use the dashboard as a decision-support tool.** Forecasts, error metrics, uncertainty ranges, and capacity-risk indicators can be reviewed together rather than in isolation.
6. **Retrain and recalibrate periodically.** Operational conditions can change over time, so model performance and residual behavior should be monitored before relying on forecasts for high-impact planning.

8. Limitations

The dataset contains substantial and systematic missing dates, limiting direct interpretation as a complete daily time series. Longer-horizon errors are substantially higher than short-horizon errors. Capacity-risk estimates are retrospective and based on a scenario threshold of 3,000; they should not be interpreted as validated operational probabilities. Additional external drivers and more complete reporting data could improve future forecasts.

9. Conclusion

The project demonstrates that historical operational data can support useful short-term forecasting of HHS care load. Gradient Boosting substantially outperformed the tested naive and statistical baselines in the final evaluation, with the strongest performance at 1-day and 7-day horizons. The most important practical finding is that forecast value depends not only on predictive accuracy, but also on data quality, uncertainty communication, and appropriate use of different forecast horizons. The resulting Streamlit dashboard translates these analytical outputs into an accessible decision-support format for operational and planning stakeholders.